

# Automating Daily Stock Dump Processing

## with Excel VBA

### PROJECT OBJECTIVE

Automate the validation and enrichment of distributor stock dump files using a centralized State Master mapping — reducing repetitive manual effort and ensuring consistent, reliable outputs.

...

```
Sub ProcessDumps ()  
  
    Dim dict As Dictionary  
    Dim wbDump As Workbook  
  
    LoadStateMaster dict  
  
    For Each f In folder  
        ValidateERP f, dict  
        AddFinalState f  
        RemoveInvalid f  
        SaveOutput f  
    Next f  
  
End Sub
```

## Why Was This Problem Worth Solving?



Daily stock dump files were generated and stored separately for each day



Each file contained distributor-level inventory data with ERP IDs



ERP IDs needed to be mapped with state information from a master reference file



Unmapped distributors had to be identified and excluded from reporting



Performing this manually for hundreds of files was repetitive and error-prone

### IMPACT

#### High Manual Effort

Repeated steps per file

#### Inconsistent Outputs

Human error risk

#### Difficult to Scale

Historical data processing

#### No Audit Trail

Manual work leaves gaps

## Before Automation vs After Automation

### BEFORE AUTOMATION

→ Open File



→ Apply Lookup



→ Validate ERP IDs



→ Remove Invalid Records



→ Save Output



↻ Repeat for Each File

VS

### AFTER AUTOMATION

→ Run VBA Macro



→ Read Master Mapping



→ Process All Files



→ Validate ERP IDs

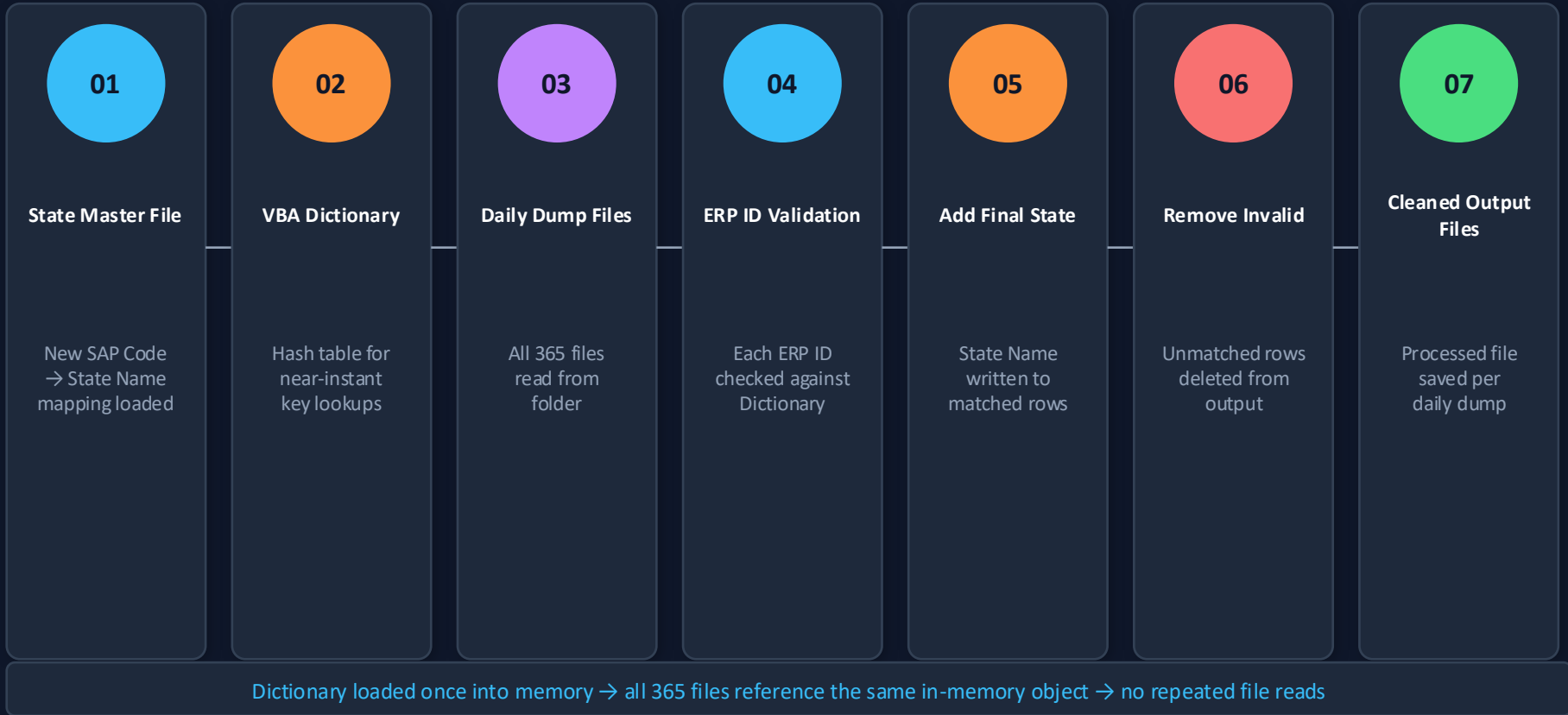


→ Generate Cleaned Outputs



✓ Done

# VBA Automation Workflow



# What I Built



## VBA Batch Processing Framework

Dynamically reads and loops through all files in a folder without manual selection



## Dictionary-Based Lookup Engine

Scripting.Dictionary loads State Master once, enabling instant ERP ID validation per row



## Array-Driven Data Handling

Worksheet data loaded into arrays for bulk operations — avoids slow cell-by-cell reads



## Automated Output Generation

Cleaned and enriched workbooks saved automatically with consistent naming

### VBA CONCEPTS APPLIED

- ✓ Workbook & Worksheet Objects
- ✓ Scripting.Dictionary
- ✓ Arrays for Performance
- ✓ File Handling using Dir() Function
- ✓ For Each / For Next Loops
- ✓ Conditional Logic (If/Then)
- ✓ Dynamic Range Operations

## Business Context & Technical Achievement

**365**

Files Processed

**~500+**

Master Records

**500K+**

Rows per File (avg)

**365**

Cleaned Outputs

### BUSINESS CONTEXT

- ▶ Daily stock dump files received from operations
- ▶ Distributor ERP IDs required state mapping from master data
- ▶ Invalid ERP IDs needed to be excluded from reporting
- ▶ Process needed to be consistent and repeatable across all files

### TECHNICAL ACHIEVEMENT

- ✓ Automated batch processing of 365 daily stock dump files
- ✓ Dictionary-based ERP ID validation — fast, memory-efficient lookups
- ✓ Dynamic state mapping enriched from centralized master data
- ✓ Cleaned output files generated automatically for each processed dump

# What This Project Taught Me



VBA can automate large-scale Excel operations effectively without any external tools



Dictionaries provide significantly faster lookups than traditional worksheet formulas like VLOOKUP



Arrays improve performance drastically when handling large datasets — fewer sheet interactions



Batch processing reduces repetitive manual work and creates a reusable framework for future runs



Automation enables analysts to focus on insights and decision-making rather than data preparation

## SKILLS DEMONSTRATED

◆ Excel VBA

◆ Data Cleaning

◆ Data Transformation

◆ Data Validation

◆ Master Data Mapping

◆ Batch Processing

◆ Process Automation

◆ File Handling

◆ Lookup Optimization

◆ Reporting Automation

## Where This Can Go Next



### Power Query Migration

Build a Power Query version capable of handling even larger datasets with native refresh support



### Automated Logging & Exception Reporting

Add a processing log sheet capturing file name, record count, matched vs unmatched rows per run



### User-Friendly Interface

Build a simple VBA UserForm so non-technical users can run the macro without opening the code



### Windows Task Scheduler Integration

Schedule daily automated processing via Task Scheduler to run without any manual trigger



### Processing Summary Dashboard

Auto-generate a summary sheet showing daily processing stats, file counts, and exception flags



### Python / Power Automate Version

Re-architect using Python pandas or Power Automate for cloud-based scalable execution